$$
\begin{aligned}
&C_{\phi l}^{(12)[3]i1\_i2\_} \to \frac{1}{16\pi^2}\Bigg(\frac{1}{3} m_2 \tilde{\mu} g_2^4 \, \mathrm{LF}_{2,2,0}[m_2, \tilde{\mu}] \, \delta_{i1i2} - \frac{1}{6} m_2 \tilde{\mu} g_2^4 \, \mathrm{LF}_{3,2,-1}[m_2, \tilde{\mu}] \, \delta_{i1i2} + \\
&\frac{1}{8} \tilde{\mu} g_2^2 \, \overline{y_e}^{pr} a_e^{pr} \, \mathrm{LF}_{4,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] \delta_{i1i2} - \frac{1}{6} \tilde{\mu} g_2^2 \, \overline{y_e}^{pr} a_e^{pr} \, \mathrm{LF}_{5,1,-2}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] \delta_{i1i2} + \\
&\frac{3}{8} \tilde{\mu} g_2^2 \, \overline{y_d}^{pr} a_d^{pr} \, \mathrm{LF}_{4,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] \delta_{i1i2} - \frac{1}{2} \tilde{\mu} g_2^2 \, \overline{y_d}^{pr} a_d^{pr} \, \mathrm{LF}_{5,1,-2}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] \delta_{i1i2} + \\
&\frac{1}{4} \tilde{\mu} g_2^2 \, \overline{y_u}^{pr} a_u^{pr} \, \mathrm{LF}_{2,2,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] \delta_{i1i2} - \frac{1}{8} \tilde{\mu} g_2^2 \, \overline{y_u}^{pr} a_u^{pr} \, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] \delta_{i1i2} - \\
&\frac{1}{4} \tilde{\mu} g_2^2 \, \overline{y_u}^{pr} a_u^{pr} \, \mathrm{LF}_{3,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] \delta_{i1i2} - \frac{1}{4} \tilde{\mu} g_2^2 \, \overline{y_u}^{pr} a_u^{pr} \, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] \delta_{i1i2} + \\
&\frac{3}{4} \tilde{\mu} g_2^2 \, \overline{y_u}^{pr} a_u^{pr} \, \mathrm{LF}_{4,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] \delta_{i1i2} - \frac{1}{2} \tilde{\mu} g_2^2 \, \overline{y_u}^{pr} a_u^{pr} \, \mathrm{LF}_{5,1,-2}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] \delta_{i1i2} + \\
&\frac{1}{8} m_1 \tilde{\mu} g_1^2 g_2^2 \, \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_1] \, \delta_{i1i2} - \frac{1}{6} m_1 \tilde{\mu} g_1^2 g_2^2 \, \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_1] \, \delta_{i1i2} - \\
&\frac{1}{3} m_2 \tilde{\mu} g_2^4 \, \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_2] \, \delta_{i1i2} - \frac{1}{3} m_2 \tilde{\mu} g_2^4 \, \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_2] \, \delta_{i1i2} + \\
&\frac{7}{8} m_2 \tilde{\mu} g_2^4 \, \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_2] \, \delta_{i1i2} - \frac{1}{2} m_2 \tilde{\mu} g_2^4 \, \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_2] \, \delta_{i1i2} - \\
&\frac{1}{8} m_1 \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,1,1,0}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}\right] - \frac{1}{16} m_1 \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,2,1,-1}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}\right] + \\
&\frac{1}{8} m_1 \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}\right] + \frac{1}{4} m_1 \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,1,1,0}\left[m_1, m_{\tilde{e}}^{p}, \tilde{\mu}\right] - \\
&\frac{1}{4} m_1 \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{3,1,1,-1}\left[m_1, m_{\tilde{e}}^{p}, \tilde{\mu}\right] + \frac{1}{16} m_1 \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,2,1,-1}\left[m_1, m_{\tilde{l}}^{i2}, m_{\tilde{e}}^{p}\right] - \\
&\frac{3}{16} m_1 \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{e}}^{p}\right] + \frac{1}{16} m_1 \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,2,1,-1}\left[m_1, \tilde{\mu}, m_{\tilde{l}}^{i2}\right] + \\
&\frac{1}{16} m_2 \tilde{\mu} g_2^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{e}}^{p}\right] - \frac{1}{4} m_2 \tilde{\mu} g_2^4 \, \mathrm{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}\right] \delta_{i1i2} - \\
&\frac{3}{16} m_2 \tilde{\mu} g_2^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,2,1,-1}\left[m_2, \tilde{\mu}, m_{\tilde{l}}^{i2}\right] - \frac{1}{8} m_1 \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i2}\right] + \\
&\frac{1}{8} m_1 \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{l}}^{i2}\right] + \frac{3}{8} m_2 \tilde{\mu} g_2^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,1,1,0}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i2}\right] - \\
&\frac{3}{8} m_2 \tilde{\mu} g_2^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{3,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{l}}^{i2}\right] + \frac{1}{16} \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} a_e^{i1p} \, \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] - \\
&\frac{1}{32} \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} a_e^{i1p} \, \mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] + \frac{1}{16} m_1 \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} y_e^{i1p} \\
&\quad \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] + \frac{1}{32} m_1 \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,1,1,1,-1}\left[m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] + \\
&\frac{1}{16} \tilde{\mu} g_2^2 \, \overline{y_e}^{i2p} a_e^{i1p} \, \mathrm{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] - \frac{1}{32} \tilde{\mu} g_2^2 \, \overline{y_e}^{i2p} a_e^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] + \frac{1}{16} m_2 \tilde{\mu} g_2^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{1,1,1,1,0}\left[m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] + \\
&\frac{1}{32} m_2 \tilde{\mu} g_2^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,1,1,1,-1}\left[m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] - \frac{1}{16} \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} a_e^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_1, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{i2}\right] + \frac{1}{32} \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} a_e^{i1p} \, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] - \\
&\frac{1}{32} m_1 \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_1, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] + \frac{1}{16} \tilde{\mu} g_2^2 \, \overline{y_e}^{i2p} a_e^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_2, m_{\tilde{l}}^{i1}, m_{\tilde{l}}^{i2}\right] + \frac{1}{32} \tilde{\mu} g_2^2 \, \overline{y_e}^{i2p} a_e^{i1p} \, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}\right] - \\
&\frac{1}{32} m_2 \tilde{\mu} g_2^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{e}}^{p}, m_2, m_{\tilde{l}}^{i2}, \tilde{\mu}\right] - \frac{1}{32} \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} a_e^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{p}, \tilde{\mu}\right] - \frac{1}{32} \tilde{\mu} g_2^2 \, \overline{y_e}^{i2p} a_e^{i1p} \, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{l}}^{i1}, m_2, m_{\tilde{e}}^{p}, \tilde{\mu}\right] + \\
&\frac{1}{32} m_1 \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{l}}^{i2}, m_1, m_{\tilde{e}}^{p}, \tilde{\mu}\right] + \frac{1}{32} m_2 \tilde{\mu} g_2^2 \, \overline{y_e}^{i2p} y_e^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}\left[m_{\tilde{l}}^{i2}, m_2, m_{\tilde{e}}^{p}, \tilde{\mu}\right] + \frac{1}{32} \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} a_e^{i1p} \, \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}\right] - \\
&\frac{1}{32} m_1 \tilde{\mu} g_1^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_1, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}\right] + \frac{1}{32} \tilde{\mu} g_2^2 \, \overline{y_e}^{i2p} a_e^{i1p} \\
&\quad \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i1}\right] - \frac{1}{32} m_2 \tilde{\mu} g_2^2 \, \overline{y_e}^{i2p} y_e^{i1p} \, \mathrm{LF}_{2,1,1,1,-1}\left[\tilde{\mu}, m_2, m_{\tilde{e}}^{p}, m_{\tilde{l}}^{i2}\right]\Bigg)
\end{aligned}
$$